Network Intrusion Detection using a Neural Network

UE24CS352A — Machine Learning Mini-Project Write-up | Team 4 | AMITH B A - PES1UG24AM032,ASHWIN ANAND - PES1UG24AM057

1. Problem Statement

Network intrusion detection systems (IDS) must distinguish normal traffic from an increasingly diverse set of cyberattacks, while generalizing to attack variants not seen during training. This project builds a supervised Artificial Neural Network (ANN) that classifies each network connection record into one of five categories — **Normal, DoS, Probe, R2L, or U2R** — based on 41 traffic/connection features.

2. Dataset

We use the **NSL-KDD** dataset (Tavallaee et al., 2009), a cleaned revision of the KDD Cup 1999 dataset widely used as an IDS benchmark. Each record has 41 features: basic TCP/IP attributes, content features, and traffic features computed over 2-second windows. ~39 fine-grained attack names are mapped to four standard categories plus Normal. The training set is dominated by Normal/DoS records, with R2L and U2R making up a small fraction — a severe class imbalance that is the central challenge of this task.

3. Approach

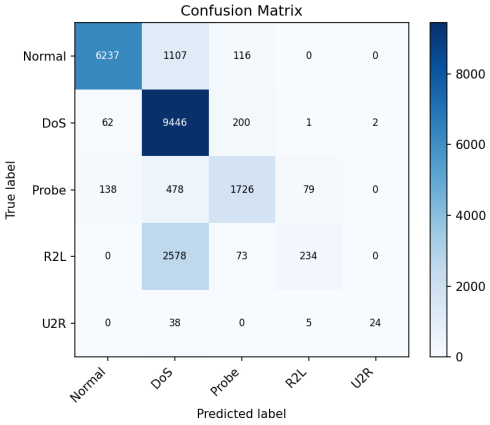
- **Preprocessing:** one-hot encode protocol_type/service/flag; standardize numeric features using training-set statistics.
- **Label mapping:** ~39 attack names → {Normal, DoS, Probe, R2L, U2R}.
- **Model:** feed-forward ANN, 2 hidden layers (128, 64 ReLU units), backend = scikit-learn MLP, Adam optimizer, L2 regularization, early stopping.
- **Evaluation:** accuracy, macro F1-score, confusion matrix, and a binary Normal-vs-Attack view.

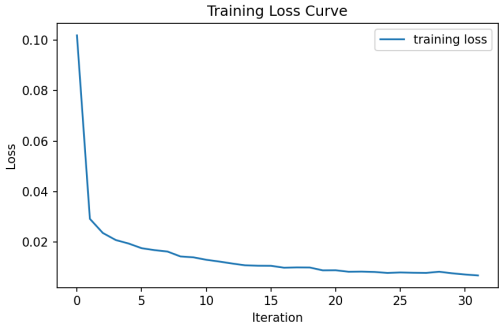
4. Implementation Overview

Implemented in Python (pandas, scikit-learn/TensorFlow, matplotlib) as a single reproducible notebook: load NSL-KDD → map labels → one-hot encode + scale → train the ANN → evaluate → plot. After one-hot encoding, the 41 raw features expand to 122 input dimensions.

5. Results

Metric	Value
Train samples / Test samples	125,973 / 22,544
Input features (after one-hot encoding)	122
Train accuracy (multi-class)	99.8%
Test accuracy (multi-class)	78.4%
Test macro F1-score	0.63
Test accuracy (Normal vs Attack, binary view)	93.7%





As expected from the dataset's class imbalance, Normal and DoS are classified with high precision/recall, while R2L and U2R — the rarest classes — are harder to recall. This is why macro F1 and the binary Normal-vs-Attack view are reported alongside raw accuracy.

6. Conclusions

A feed-forward neural network trained on NSL-KDD's 41 engineered features reliably separates Normal traffic from the dominant attack categories, but struggles with rare R2L/U2R classes without additional handling such as class weighting or oversampling. For an operational IDS, binary Normal-vs-Attack accuracy is the most operationally relevant metric, since even a misclassified attack category still triggers an alert.